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DEALER ENQUIRIES SOLICITED

Q1. What's the difference between LCD and LED TVs?

Rafiq Sherasiya
Through e-mail

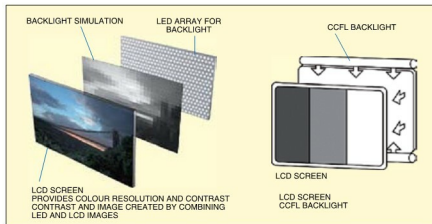
A1. Conventional TVs have a bulky cathode ray tube (CRT) screen. LCD and LED TVs are new innovations that basically belong to the same family as they both have liquid crystal display (LCD) screens.

LCDs have black segments on a light grey background, and can be as simple as one-digit display or as complex as the entire display screen of a TV. These do not emit light, so they cannot be read directly in low

advantages over standard LCD TVs that use cold-cathode fluorescent lamps (CCFL) backlighting. Comparison between LCD and LED TVs is given below.

Contrast and black levels. Compared to conventional LCD TVs, LED TVs have better contrast ratio. The backlight of LCD TVs is constantly 'on' during use. Dark areas in LCD TVs are created by turning the light away from the LCD panel, resulting in a low contrast level and poor clarity in dark scenes.

In LED TVs, light-emitting diodes are dimmed locally to create deeper



LCD TV screen with LED backlight (L) and with CCFL backlight (R)

ambient light. For applications in such surroundings, a backlight is provided. Backlights illuminate the LCD from the side or back of the display panel.

The main difference between LED and LCD TVs therefore lies in the technique used for backlighting. The screen remains the same but techniques for backlighting are different. LCD TVs use some form of fluorescent lighting from tubes to make more advanced flat arrays of lights, whereas LED TVs use light-emitting diodes to light up the LCD panel. Just as there are different styles of fluorescent lights in traditional LCDs, there are also different styles of LED backlighting. The two arrangements are shown in Figs 1 and 2.

Though LED TVs are essentially a form of LCD TVs, these have some ad-

black in specific areas of the screen, which allows for more detail in dark scenes.

Colour accuracy. There is not much difference in the colour quality of both the TVs. However, LED TVs, with the RGB colour light technology, or a colour wheel to affect the backlight colour, can manipulate the backlight to capture accurate colours of the image.

Viewing angles. While useable viewing angle of traditional LCDs has improved over the years, this is still a shortcoming and even the best of LCDs suffer from contrast degradation when viewed from angles wider than around 30 degrees off centre. LED-backlit LCDs overcome this problem and can match the viewing angle of expensive plasma TVs.

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